

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	28 June 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Import electricity consumption dataset (Jan 2019–Dec 2020).
FR-2	Data Preparation	Clean missing values and standardize state and date formats.
FR-3	Trend Analysis	Analyse month-wise electricity consumption trends.
FR-4	Year-wise Comparison	Compare electricity consumption between 2019 and 2020.
FR-5	Regional Analysis	Group states into regions and compare electricity demand.
FR-6	Lockdown Impact Analysis	Analyse electricity consumption during March–June 2020.
FR-7	Recovery Analysis	Measure post-lockdown recovery across states.
FR-8	Dashboard Development	Create interactive charts and filters.
FR-9	Reporting	Generate insights and recommendations for policymakers.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Performance	Dashboard should load and display visualizations quickly.
NFR-2	Accuracy	Analysis and calculations should
NFR-3	Usability	Dashboard should be easy to understand and navigate.
NFR-4	Scalability	Solution should support additional years of data in the future.
NFR-5	Availability	Dashboard and reports should be accessible whenever needed.
NFR-6	Reliability	System should produce consistent results with the same data.
NFR-7	Maintainability	Dashboards and calculations should be easy to update.
NFR-8	Security	Dataset should be stored and handled securely.